ACTON
RESEARCH CORPORATION
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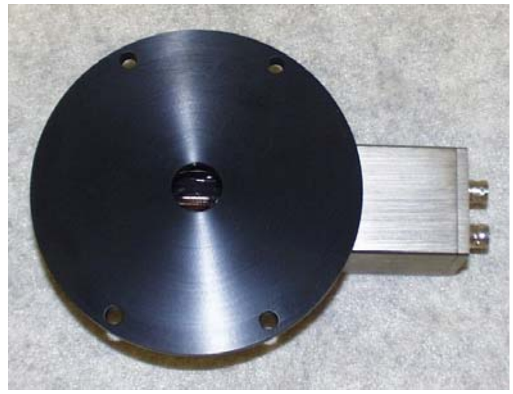
Phone: (978)263-3584, **Fax:** (978)263-5086
Web Site: www.acton-research.com
www.piacton.com

DA-780 Side Window Detector Assembly

Serial Number:

Ver. 2.0



The Model DA-780 is a detector assembly for use with ARC and other commercial monochromators with a 5-inch diameter slit housing. The DA-780 detector housing is nickel plated to assure proper grounding, and is designed to accept 1-1/8 inch diameter side window photomultiplier tubes. Connectors are provided on the bottom of the housing for connecting the high voltage (MHV) and signal (BNC) cables. The DA-780 is available with several different PM tubes as listed below. Other tubes are available on request.

DA-780-VUV:

The DA-780-VUV is for vacuum use over the wavelength range from 30nm to 600nm. It utilizes a vacuum sealed, sodium-salicylate-coated window as a wavelength converter in front of a photomultiplier tube, such as an EMI 9781B with an extended S5 response or equivalent. The window is easily removable for re-coating if necessary.

DA-780-UV:

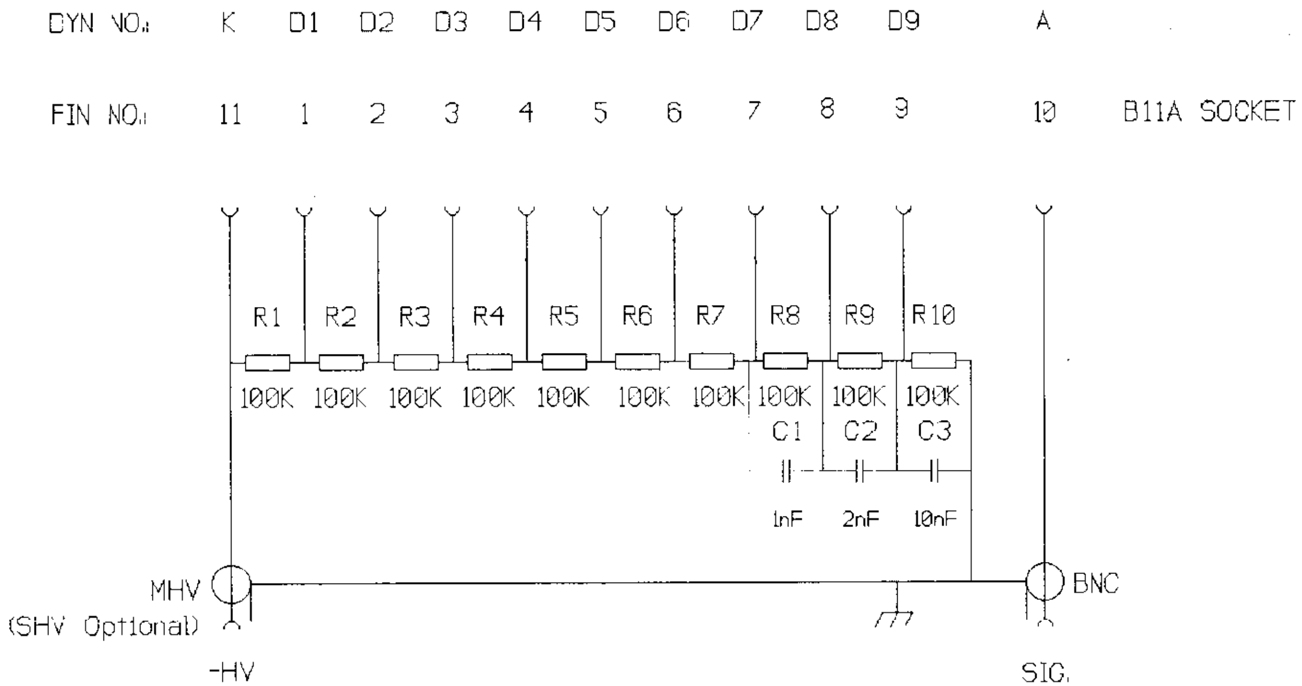
The DA-780-UV is for atmospheric use over the range 180nm to 700nm and is supplied with an EMI 9783B tube or equivalent (e.g., Hamamatsu R928 PMT).

DA-780-IR:

The DA-780-IR is for atmospheric use over the range of 400nm to 1100nm. A Hamamatsu tube R5108 with a response similar to S1 or an equivalent is supplied.

Electron Tubes Limited
Bury Street Ruislip
Middx HA4 7TA UK

electron tubes



NOTES: Resistors 0.25W Metal Film 1% Tol.

Capacitors 500V Disc Ceramic 10% Tol.

Maximum Voltage 1250V Subject To Not Exceeding Tube Max. A/lm

Tube Types 9780, 9781, 9783, 9785

Issue	Date	C/Note No.	Sign

Drawn by	DCF
Date	24-5-95
Scale	n/a
Checked	
Approved	

Project: SWH Series Housings

Title: Voltage Divider For Side Window Tubes

Drawing Number: 25823 D59

F/EB10 issue 2

ACCESSORIES

30 mm (1⅛″) photomultiplier
9781B side window series data sheet

electron tubes

1 description

The 9781B is a 30 mm (1⅛″) diameter, side window photomultiplier with a blue-green sensitive bialkali photocathode and 9 high gain, high stability, SbCs dynodes of circular focused design.

2 applications

- wide range of applications
- spectroscopy
- high light level

3 features

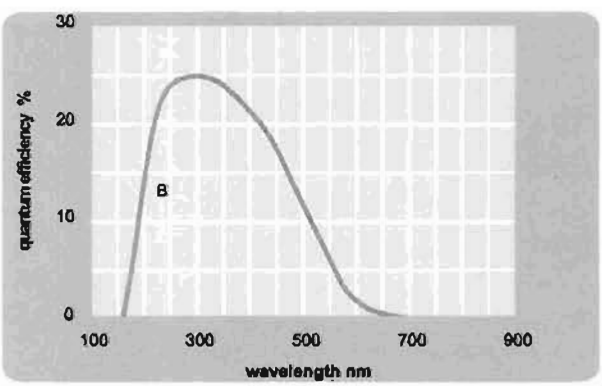
- compact
- low operating voltage
- UV sensitive

4 window characteristics

	9781B UV glass
spectral range (nm)*	165 - 680
refractive index (n_d)	1.48
K (ppm)	8500
Th (ppb)	30
U (ppb)	30

* wavelength range over which quantum efficiency exceeds 1 % of peak

5 typical spectral response curves

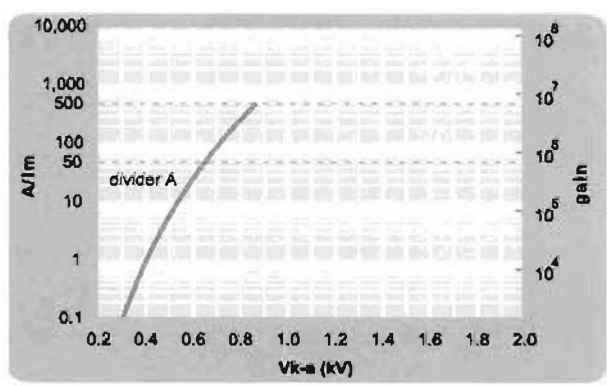


6 characteristics

	unit	min	typ	max
photocathode: bialkali				
active diameter	mm		8 x 24	
quantum efficiency at peak	%		25	
luminous sensitivity	µA/lm		75	
with CB filter		5	10	
with CR filter			5	
dynodes: 9CFSbCs				
anode sensitivity in divider A:				
nominal anode sensitivity	A/lm		50	
max. rated anode sensitivity	A/lm		500	
overall V for nominal A/lm	V		650	800
overall V for max. rated A/lm	V		850	
gain at nominal A/lm	$x\ 10^6$		0.7	
dark current at 20 °C:				
dc at nominal A/lm	nA		0.2	5
dc at max. rated A/lm	nA		2	
pulsed linearity (-5% deviation):				
divider A	mA		10	
rate effect (I_a for $\Delta g/g$=1%):	µA		20	
magnetic field sensitivity:				
the field for which the output decreases by 50 %				
most sensitive direction	$T\ x\ 10^{-4}$		25	
temperature coefficient:	% $°C^{-1}$		±0.5	
timing:				
single electron rise time	ns		2	
single electron (fwhm)	ns		4	
single electron jitter (fwhm)	ns		2.2	
transit time	ns		20	
weight:	g		45	
maximum ratings:				
anode current	µA			100
cathode current	nA			5000
gain	$x10^6$			7
sensitivity	A/lm			500
temperature	°C	-80		60
V (k-a)(1)	V			1250
V (k-d1)	V			150
V (d-d)(2)	V			150
ambient pressure (absolute):	kPa			202

(1) subject to not exceeding max. rated sensitivity (2) subject to not exceeding max rated V(k-a)

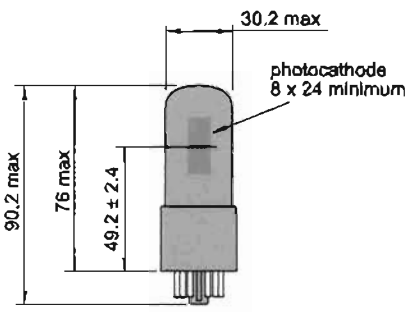
7 typical voltage gain characteristics



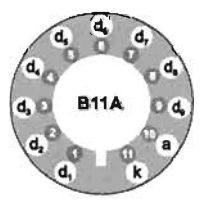
8 voltage divider distribution (type A)

	k	d1	d2	···	d8	d9	a	
	A	R	R	···················	R	R		Standard

9 external dimensions mm



10 base configuration (viewed from below)



The B11A socket range, available for this series, includes versions with or without a mounting flange, and with contacts for mounting directly onto printed circuit boards.

11 ordering information

The 9781B is the parent type. It meets the published specification contained in this data sheet. Variants are listed below with the convention for deriving the type number that includes your chosen selection(s). For one-off requirements the selection will change the B suffix to A, or for ongoing requirements Electron Tubes will advise a 2 digit suffix that maintains the customer's requirements.

9781 ▢ ▢

selection options
M supplied with spectral response calibration

specification
B as data sheet
A customer specific selection(s) - single order
Bnn customer specific selection(s) - repeat order

12 voltage dividers

The standard voltage dividers available for these pmts are tabulated below:

	k	d1	d2	d3 ···	d8	d9	a
C639A	R	R	R	····················	R	R	
C699	V	V	V	····················	V	V	

R = 100 kΩ

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DS_ 9781B Series Issue 1
01 June 2001

30 mm (1⅛″) photomultiplier
9783B side window series data sheet

electron tubes

1 description

The 9783B is a 30 mm (1⅛″) diameter, side window photomultiplier with a blue-green sensitive bialkali photocathode and 9 high gain, high stability, SbCs dynodes of circular focused design.

2 applications

- wide range of applications
- spectroscopy
- high light level

3 features

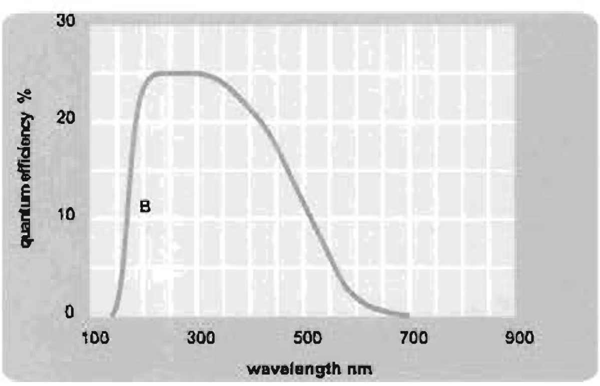
- compact
- low operating voltage
- extended UV sensitivity

4 window characteristics

	9783B fused silica
spectral range (nm)*	160 - 680
refractive index (n_d)	1.46
K (ppm)	<10
Th (ppb)	<10
U (ppb)	<10

* (wavelength range over which quantum efficiency exceeds 1 % of peak)

5 typical spectral response curves

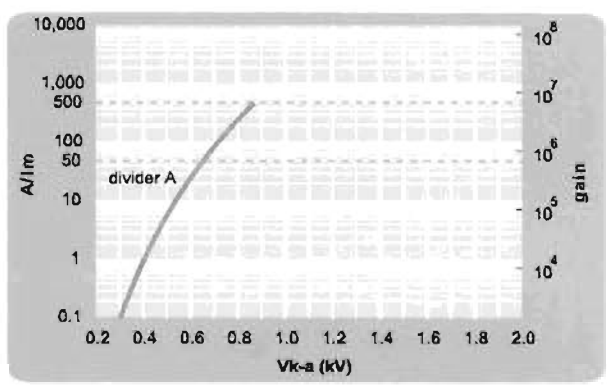


6 characteristics

	unit	min	typ	max
photocathode: bialkali				
active diameter	mm		8 x 24	
quantum efficiency at peak	%		25	
luminous sensitivity	µA/lm		75	
with CB filter		5	10	
with CR filter			5	
dynodes: 9CFSbCs				
anode sensitivity in divider A:				
nominal anode sensitivity	A/lm		50	
max. rated anode sensitivity	A/lm		500	
overall V for nominal A/lm	V		650	800
overall V for max. rated A/lm	V		850	
gain at nominal A/lm	$\times 10^6$		0.7	
dark current at 20 °C:				
dc at nominal A/lm	nA		0.5	5
dc at max. rated A/lm	nA		5	
pulsed linearity (-5% deviation):				
divider A	mA		10	
rate effect (I_a for $\Delta g/g$=1%):	µA		20	
magnetic field sensitivity:				
the field for which the output decreases by 50 %				
most sensitive direction	$T \times 10^{-4}$		25	
temperature coefficient:	% °C^{-1}		±0.5	
timing:				
single electron rise time	ns		2	
single electron (fwhm)	ns		4	
single electron jitter (fwhm)	ns		2.2	
transit time	ns		20	
weight:	g		45	
maximum ratings:				
anode current	µA			100
cathode current	nA			5000
gain	$\times 10^6$			7
sensitivity	A/lm			500
temperature	°C	-80		60
V (k-a)[1]	V			1250
V (k-d1)	V			150
V (d-d)[2]	V			150
ambient pressure (absolute):	kPa			202

[1] subject to not exceeding max. rated sensitivity [2] subject to not exceeding max rated V(k-a)

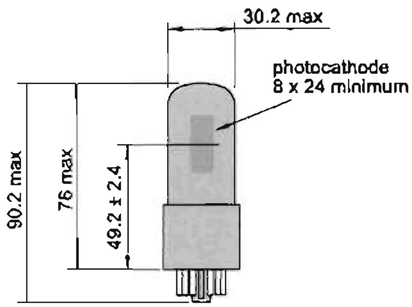
7 typical voltage gain characteristics



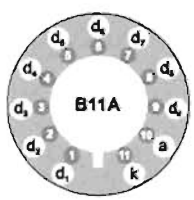
8 voltage divider distribution (type A)

k	d_1	d_2	....	d_8	d_9	a	
A	R	R	....	R	R		Standard

9 external dimensions mm



10 base configuration (viewed from below)



The B11A socket range, available for this series, includes versions with or without a mounting flange, and with contacts for mounting directly onto printed circuit boards.

11 ordering information

The 9783B is the parent type. It meets the specification contained in this data sheet. Variants are listed below with the convention for deriving the type number that includes your chosen selection(s). For one-off requirements, the selection will change the B suffix to A, or for ongoing requirements Electron Tubes will advise a 2 digit suffix after the letter B that maintains the customer's specific requirements.

9783 ■ ■

selection options
M supplied with spectral response calibration

specification
B as data sheet
A customer specific selection(s) - single order
Bnn customer specific selection(s) - repeat order

12 voltage dividers

The standard voltage dividers available for these pmts are tabulated below:

	k	d_1	d_2	d_3	....	d_8	d_9	a
C639A	R	R	R		....	R	R	
C699	V	V	V		....	V	V	

R = 100 kΩ

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HAMAMATSU

PHOTOMULTIPLIER TUBES
R928, R955

Extended Red, High Sensitivity, Multialkali Photocathode
28mm (1-1/8 Inch) Diameter, 9-Stage, Side-On

FEATURES

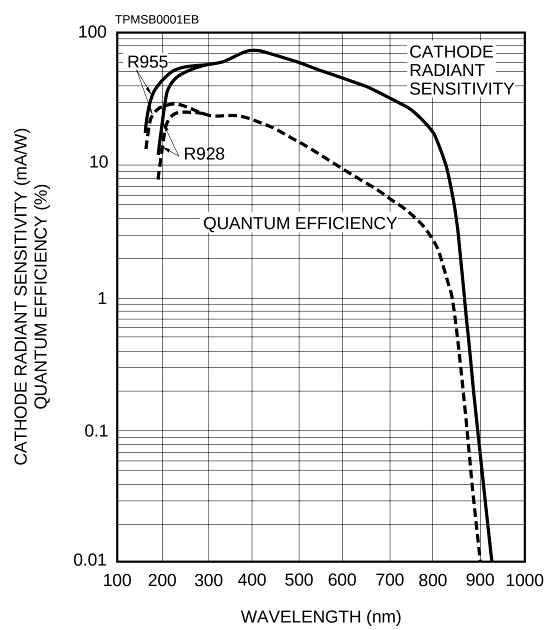
- **Wide Spectral Response**
  - **R928 .... 185 to 900 nm**
  - **R955 .... 160 to 900 nm**
- **High Cathode Sensitivity**
  - **Luminous .... 250 μ A/lm**
  - **Radiant at 400nm 74mA/W**
- **High Anode Sensitivity (at 1000V)**
  - **Luminous 2500A/lm**
  - **Radiant at 400nm .... 7.4×10^5 A/W**
- **Low Drift and Hysteresis**

The R928 and R955 feature extremely high quantum efficiency, high current amplification, good S/N ratio and wide spectral response from UV to near infrared. The R928 employs a UV glass envelope and the R955 has a fused silica envelope for UV sensitivity extension.
The R928 and R955 are well suited for use in broad-band spectrophotometers, atomic absorption spectrophotometers, emission spectrophotometers and other precision photometric instruments.

GENERAL

Parameter	Description/Value	Unit
Spectral Response R928 R955	 185 to 900 160 to 900	 nm nm
Wavelength of Maximum Response	400	nm
Photocathode Material Minimum Effective Area	 Multialkali 8 × 24	 – mm
Window Material R928 R955	 UV glass Fused silica	–
Dynode Secondary Emitting Surface Structure Number of Stages	 Multialkali Circular-cage 9	 – – –
Direct Interelectrode Capacitances Anode to Last Dynode Anode to All Other Electrodes	 Approx. 4 Approx. 6	 pF pF
Base	11-pin base JEDEC No. B11-88	–
Weight	Approx. 45	g
Suitable Socket	E678–11A (option)	–
Suitable Socket Assembly	E717–21 (option)	–

Figure 1: Typical Spectral Response



PHOTOMULTIPLIER TUBES R928, R955

MAXIMUM RATINGS (Absolute Maximum Values)

Parameter	Value	Unit
Supply Voltage		
Between Anode and Cathode	1250	Vdc
Between Anode and Last Dynode	250	Vdc
Average Anode Current	0.1	mA
Ambient Temperature	–80 to +50	℃

CHARACTERISTICS (at 25℃)

Parameter	R928 Min.	R928 Typ.	R928 Max.	R955 Min.	R955 Typ.	R955 Max.	Unit
Cathode Sensitivity							
Quantum Efficiency at Peak Wavelength	—	25.4 (at 260nm)	—	—	29.0 (at 220nm)	—	%
Luminous B	140	250	—	140	250	—	μA/lm
Radiant at 194nm	—	18	—	—	43	—	mA/W
254nm	—	52	—	—	56	—	mA/W
400nm	—	74	—	—	74	—	mA/W
633nm	—	41	—	—	41	—	mA/W
852nm	—	3.5	—	—	3.5	—	mA/W
Red/White Ratio C	0.2	0.3	—	0.2	0.3	—	—
Blue D	—	8	—	—	8	—	μA/lm-b
Anode Sensitivity							
Luminous E	400	2500	—	400	2500	—	A/lm
Radiant at 194nm	—	1.8×10^5	—	—	4.3×10^5	—	A/W
254nm	—	5.2×10^5	—	—	5.6×10^5	—	A/W
400nm	—	7.4×10^5	—	—	7.4×10^5	—	A/W
633nm	—	4.1×10^5	—	—	4.1×10^5	—	A/W
852nm	—	3.5×10^4	—	—	3.5×10^4	—	A/W
Gain E	—	1.0×10^7	—	—	1.0×10^7	—	—
Anode Dark Current F							
After 30 minute Storage in the darkness	—	3	50	—	3	50	nA
ENI(Equivalent Noise Input) H	—	1.3×10^{-16}	—	—	1.3×10^{-16}	—	W
Time Response E							
Anode Pulse Rise Time I	—	2.2	—	—	2.2	—	ns
Electron Transit Time J	—	22	—	—	22	—	ns
Transit Time Spread (TTS) K	—	1.2	—	—	1.2	—	ns
Anode Current Stability L							
Current Hysteresis	—	0.1	—	—	0.1	—	%
Voltage Hysteresis	—	1.0	—	—	1.0	—	%

NOTES

A: Averaged over any interval of 30 seconds maximum.

B: The light source is a tungsten filament lamp operated at a distribution temperature of 2856K. Supply voltage is 100 volts between the cathode and all other electrodes connected together as anode.

C: Red/White ratio is the quotient of the cathode current measured using a red filter(Toshiba R-68) interposed between the light source and the tube by the cathode current measured with the filter removed under the same conditions as Note B.

D: The value is cathode output current when a blue filter(Corning CS-5-58 polished to 1/2 stock thickness) is interposed between the light source and the tube under the same condition as Note B.

E: Measured with the same light source as Note B and with the voltage distribution ratio shown in Table 1 below.

Table 1:Voltage Distribution Ratio

Electrode	K		Dy1		Dy2		Dy3		Dy4		Dy5		Dy6		Dy7		Dy8		Dy9		P
Distribution Ratio		1		1		1		1		1		1		1		1		1		1	

Supply Voltage : 1000Vdc
K : Cathode, Dy : Dynode, P : Anode

F: Measured with the same supply voltage and voltage distribution ratio as Note E after removal of light.

G: Measured at a supply voltage adjusted to provide an anode sensitivity of 100 A/lm.

H: ENI is an indication of the photon-limited signal-to-noise ratio. It refers to the amount of light in watts to produce a signal-to-noise ratio of unity in the output of a photomultiplier tube.

$$\mathrm{ENI} = \frac{\sqrt{2q \cdot Idb \cdot G \cdot \Delta f}}{S}$$

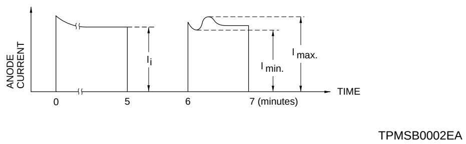
where q = Electronic charge (1.60×10^{-19} coulomb).
Idb = Anode dark current(after 30 minute storage) in amperes.
G = Gain.
Δf = Bandwidth of the system in hertz. 1 hertz is used.
S = Anode radiant sensitivity in amperes per watt at the wavelength of peak response.

I: The rise time is the time for the output pulse to rise from 10% to 90% of the peak amplitude when the entire photocathode is illuminated by a delta function light pulse.

J: The electron transit time is the interval between the arrival of delta function light pulse at the entrance window of the tube and the time when the anode output reaches the peak amplitude. In measurement, the whole photocathode is illuminated.

K: Also called transit time jitter. This is the fluctuation in electron transit time between individual pulses in the signal photoelectron mode, and may be defined as the FWHM of the frequency distribution of electron transit times.

L: Hysteresis is temporary instability in anode current after light and voltage are applied.



$$\text{Hysteresis} = \frac{I_{max.} - I_{min.}}{I_i} \times 100(\%)$$

(1)Current Hysteresis
The tube is operated at 750 volts with an anode current of 1 micro-ampere for 5 minutes. The light is then removed from the tube for a minute. The tube is then re-illuminated by the previous light level for a minute to measure the variation.

(2)Voltage Hysteresis
The tube is operated at 300 volts with an anode current of 0.1 micro-ampere for 5 minutes. The light is then removed from the tube and the supply voltage is quickly increased to 800 volts. After a minute, the supply voltage is then reduced to the previous value and the tube is re-illuminated for a minute to measure the variation.

Figure 2: Anode Luminous Sensitivity and Gain Characteristics

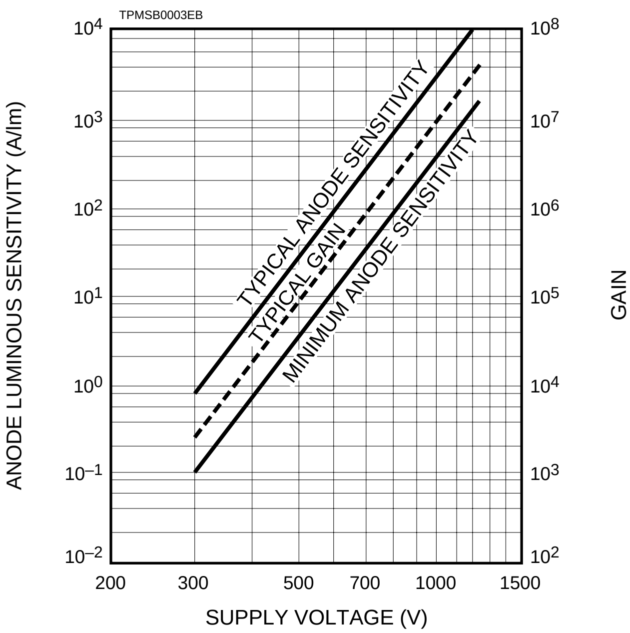


Figure 3: Typical Time Response

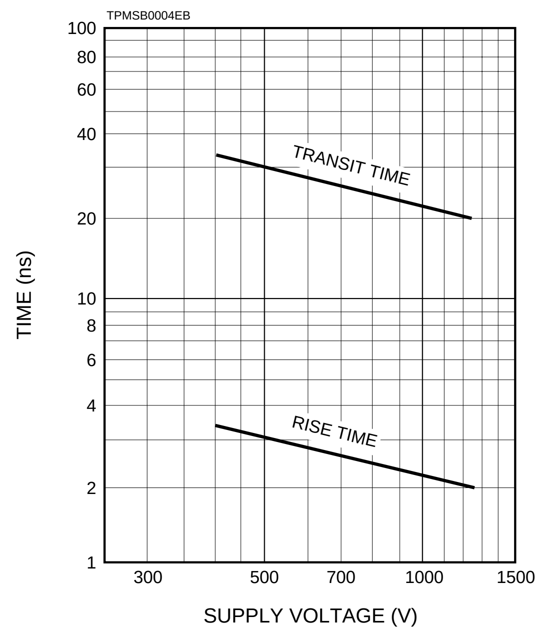


Figure 4: Typical Temperature Coefficient of Anode Sensitivity

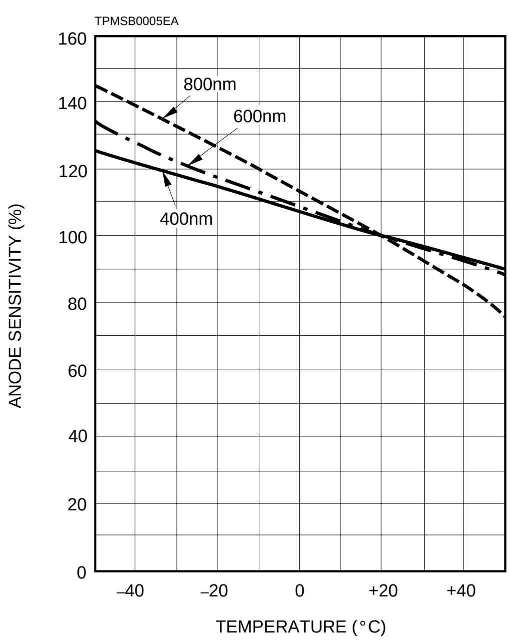


Figure 5: Typical Temperature Characteristic of Dark Current (at 1000V, after 30minute storage)

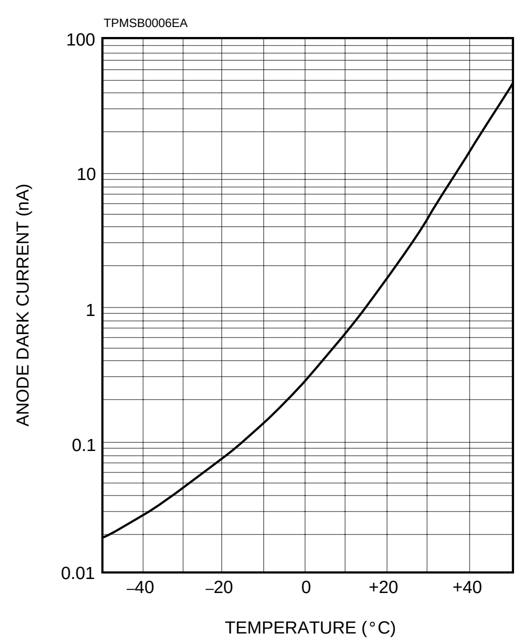
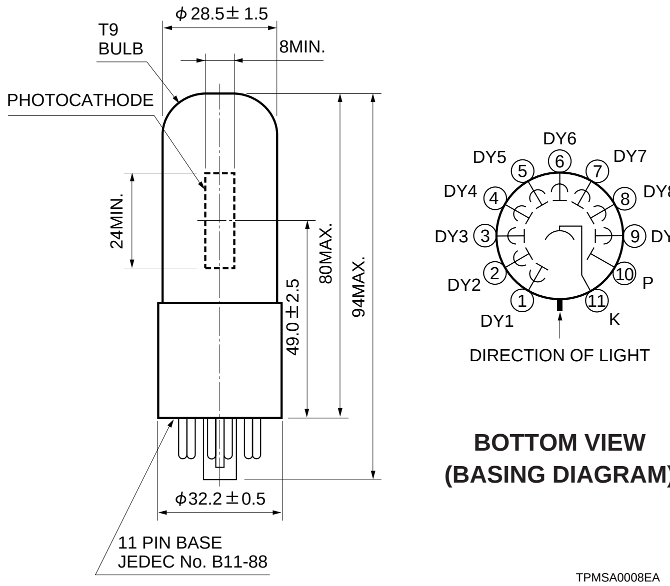


Figure 6: Dimensional Outline and Basing Diagram (Unit : mm)



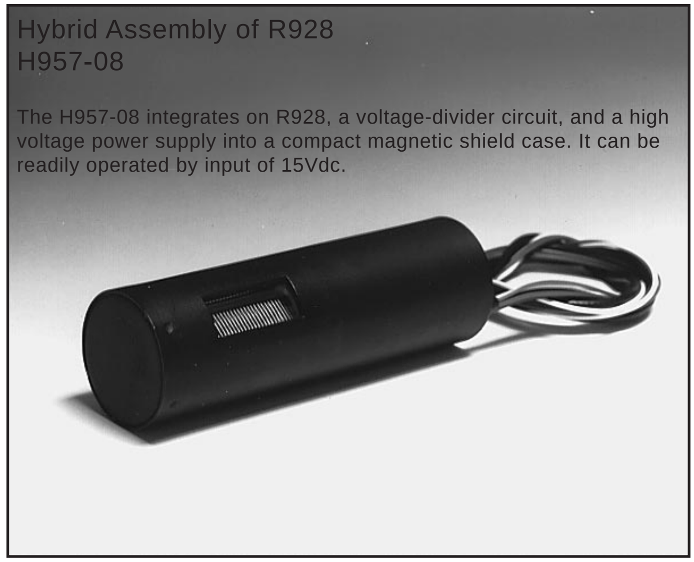
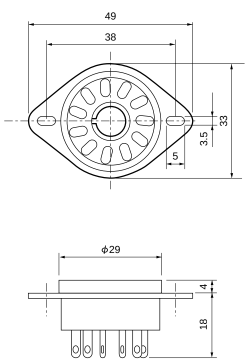
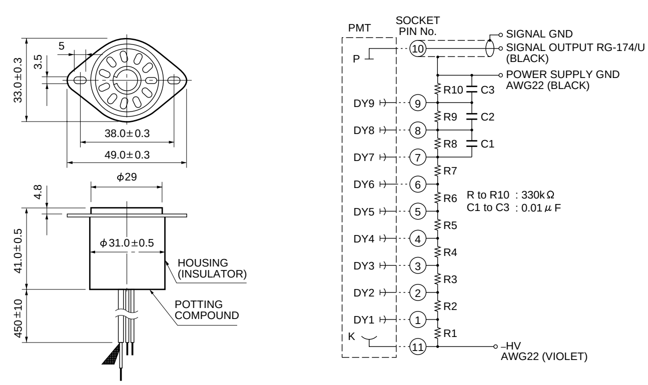


Figure 7: Optional Accessories (Unit : mm)





※Hamamatsu also provides C4900 series compact high voltage power supplies and C6270 series DP type socket assemblies which incorporate a DC to DC converter type high voltage power supply.

Warning–Personal Safety Hazards

Electrical Shock–Operating voltages applied to this device present a shock hazard.

HAMAMATSU

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314-5, Shimokanzo, Toyooka-village, Iwata-gun, Shizuoka-ken, 438-0193, Japan, Telephone: (81)539/62-5248, Fax: (81)539/62-2205
U.S.A.: Hamamatsu Corporation: 360 Foothill Road, Bridgewater. N.J. 08807-0910, U.S.A., Telephone: (1)908-231-0960, Fax: (1)908-231-1218
Germany: Hamamatsu Photonics Deutschland GmbH: Arzbergerstr. 10, D-82211 Herrsching am Ammersee, Germany, Telephone: (49)8152-375-0, Fax: (49)8152-2658
France: Hamamatsu Photonics France S.A.R.L.: 8, Rue du Saule Trapu, Parc du Moulin de Massy, 91882 Massy Cedex, France, Telephone: (33)1 69 53 71 00, Fax: (33)1 69 53 71 10
United Kingdom: Hamamatsu Photonics UK Limted: Lough Point, 2 Gladbeck Way, Windmill Hill, Enfield, Middlesex EN2 7JA, United Kingdom, Telephone: (44)181-367-3560, Fax: (44)181-367-6384
North Europe: Hamamatsu Photonics Norden AB: Färögatan 7, S-164-40 Kista Sweden, Telephone: (46)8-703-29-50, Fax: (46)8-750-58-95
Italy: Hamamatsu Photonics Italia: S.R.L.: Via Della Moia, 1/E, 20020 Arese, (Milano), Italy, Telephone: (39)2-935 81 733, Fax: (39)2-935 81 741

TPMS1001E06
MAY. 1997

HAMAMATSU

PHOTOMULTIPLIER TUBE R5108

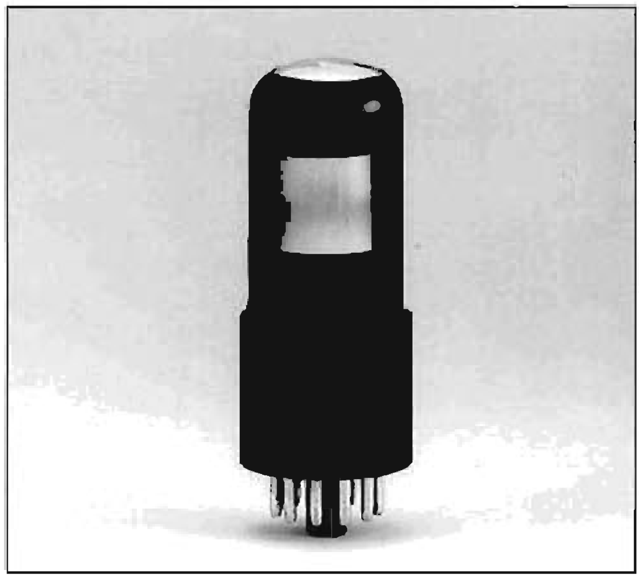
28 mm (1-1/8 Inch) Transmission Mode S–1 Photocathode, Side–on Type

FEATURES

- Wide Photocathode
- High Infrared Sensitivity
- Excellent Spatial Uniformity
- Fast Time Response

APPLICATIONS

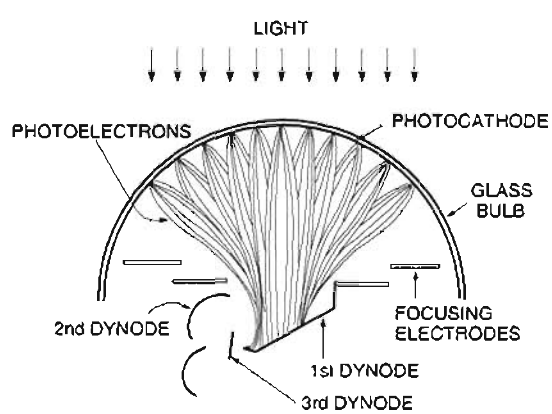
- Near Infrared Spectrophotometer
- Raman Spectrophotometer
- Photo Luminescence Measurement



GENERAL

Parameter	Description	Unit
Spectral Response	400 to 1200	nm
Wavelength of Maximum Response	800	nm
Photocathode		
Material	Ag-O-Cs	—
Minimum Effective Area (H × W)	16 × 18	mm
Window Material	Borosilicate glass	—
Dynode		
Structure	Circular-cage	—
Number of Stages	9	—
Direct Interelectrode Capacitances		
Anode to Last Dynode	1.2	pF
Anode to All Other Electrodes	3.4	pF
Base	11-pin base JEDEC No. B11-88	—
Suitable Socket	E678–11A (sold separately)	—
Applicable Socket Assembly	E717-63 (sold separately)	—
Operating Ambient Temperature	-30 to +50	°C
Storage Temperature	-30 to +50	°C

Figure 1: Electron Trajectories



PHOTOMULTIPLIER TUBE R5108

MAXIMUM RATINGS (Absolute Maximum Values)

Parameter	Value	Unit
Supply Voltage		
Between Anode and Cathode	1500	V
Between Anode and Last Dynode	250	V
Average Anode Current Ⓐ	0.01	mA

CHARACTERISTICS (at 25 °C)

Parameter	Min.	Typ.	Max.	Unit
Cathode Sensitivity				
Quantum Efficiency at 1060 nm	—	0.04	—	%
Luminous Ⓑ	10	25	—	µA/lm
Radiant at 800 nm	—	2.2	—	mA/W
Anode Sensitivity				
Luminous ⒷⒸ	3.5	7.5	—	A/lm
Gain ⒷⒸ	—	3.0×10^5	—	—
Anode Dark Current Ⓓ (after 30 min storage in the darkness)	—	350	1000	nA
Time Response Ⓒ				
Anode Pulse Rise Time Ⓔ	—	1.1	—	ns
Electron Transit Time Ⓕ	—	17	—	ns

NOTES

Ⓐ Averaged over any interval of 30 seconds maximum.

Ⓑ The light source is a tungsten filament lamp operated at a distribution temperature of 2856 K.
Supply voltage is 150 volts between the cathode and all other electrodes connected together as anode.

Ⓒ Measured with the voltage distribution ratio shown in Table 1 below.

Ⓓ Measured at the voltage which gives anode luminous sensitivity of 4 A/lm.

Ⓔ The rise time is the time for the output pulse to rise from 10 % to 90 % of the peak amplitude when the entire photocathode is illuminated by a delta function light pulse.

Ⓕ The electron transit time is the interval between the arrival of delta function light pulse at the entrance window of the tube and the time when the anode output reaches the peak amplitude. In measurement, the whole photocathode is illuminated.

Table 1: Voltage Distribution Ratio

Electrode	K	Dy1	Dy2	Dy3	Dy4	Dy5	Dy6	Dy7	Dy8	Dy9	P
Ratio	1	1	1	1	1	1	1	1	1	1	

Supply Voltage : 1250 V, K : Cathode, Dy : Dynode, P : Anode

Figure 2: Typical Spatial Uniformity

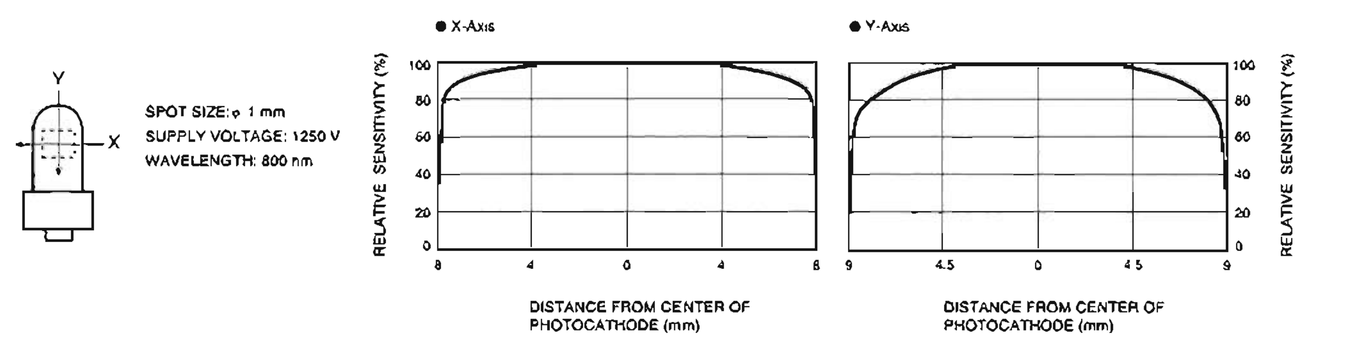


Figure 3: Typical Spectral Response

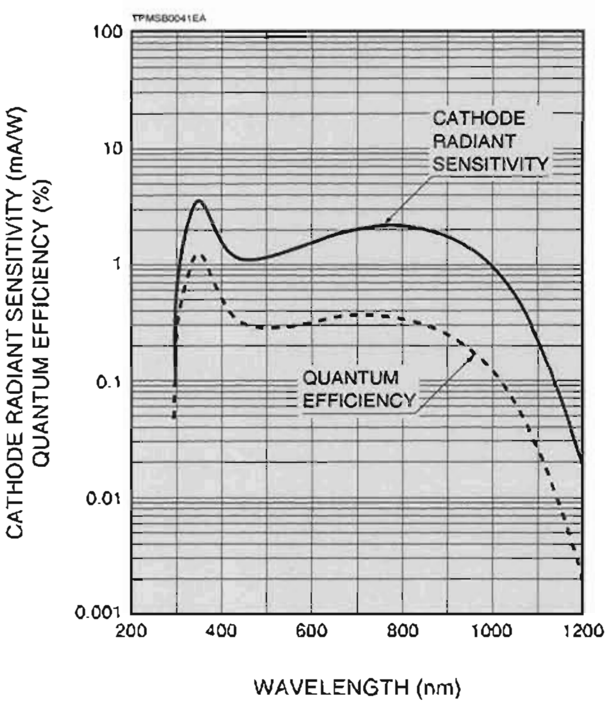


Figure 4: Typical Time Response

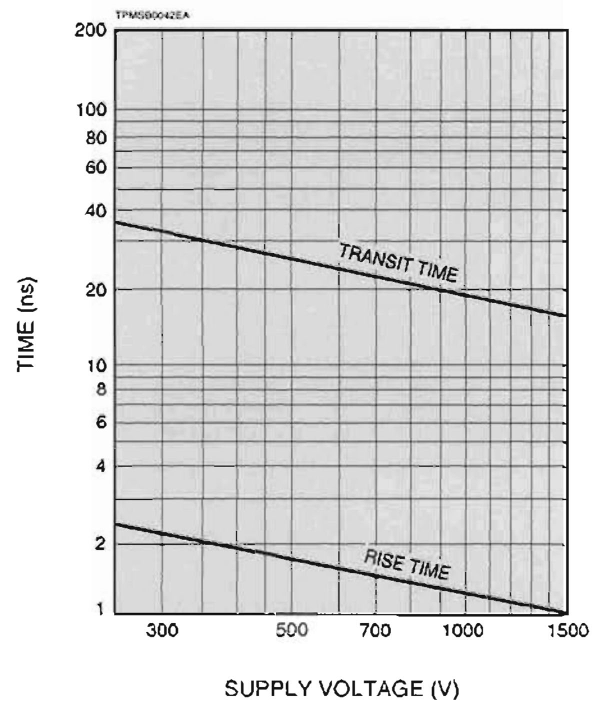


Figure 5: Typical Gain and Anode Dark Current

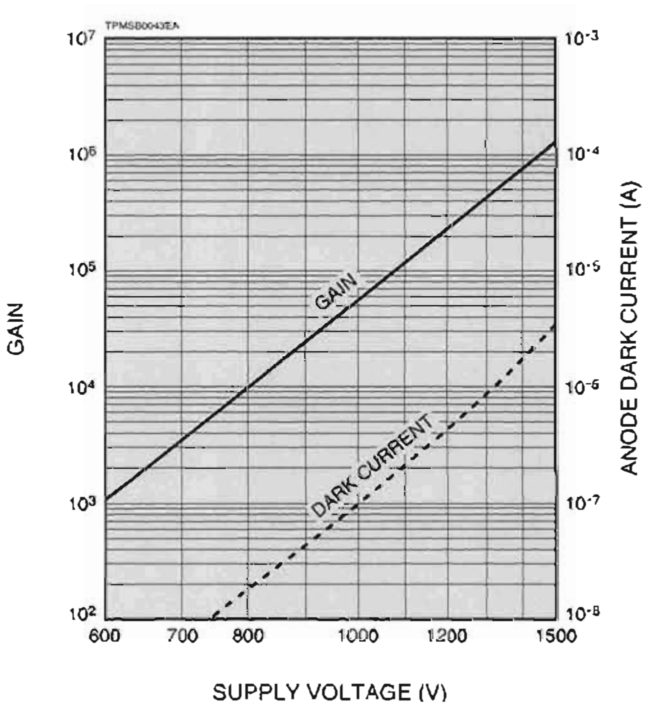


Figure 6: Typical Temperature Characteristics of Anode Dark Current (at 4 A/lm)

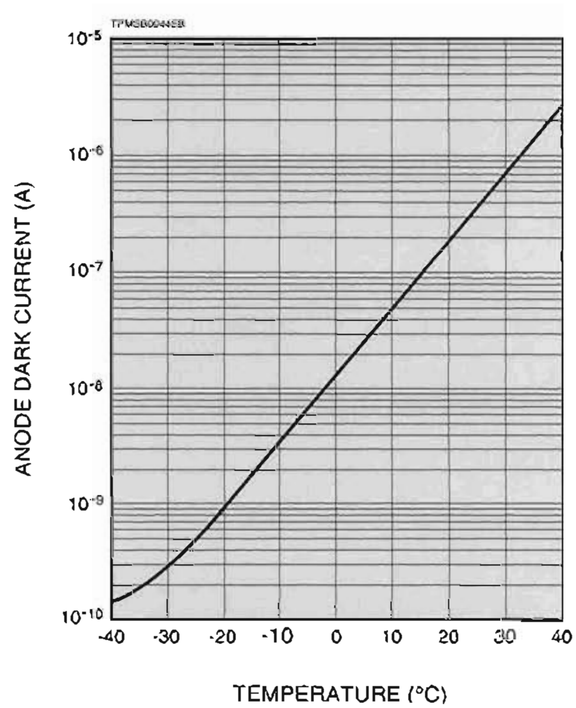


Figure 7: Dimensional Outline and Basing Diagram (Unit: mm)

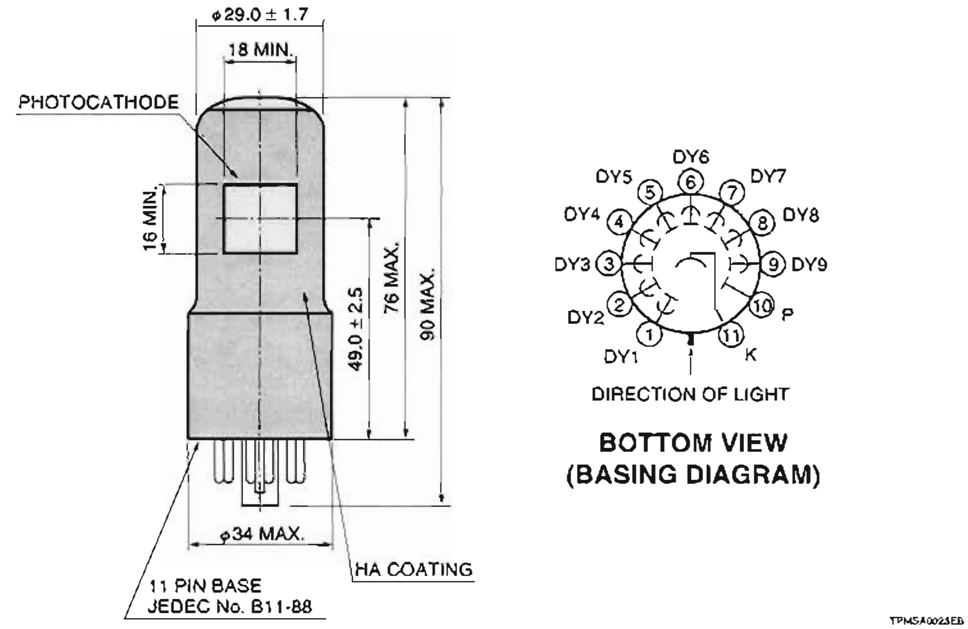
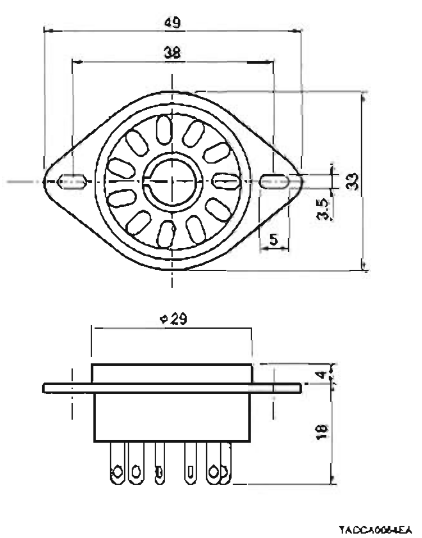
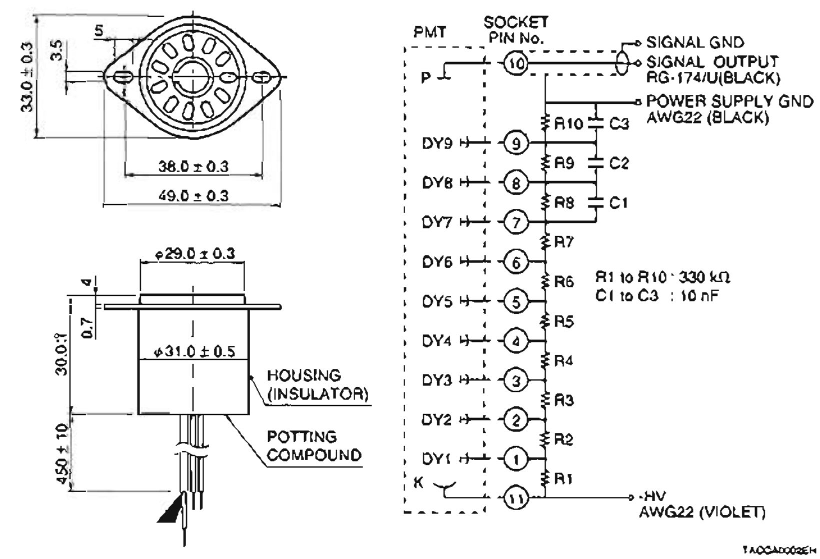


Figure 8: Optional Accessories (Unit: mm)

Socket E678-11A



D Type Socket Assembly E717-63



* Hamamatsu also provides C4900 series compact high voltage power supplies and C6270 series DP type socket assemblies which incorporate a DC to DC converter type high voltage power supply.

Warning–Personal Safety Hazards

Electrical Shock–Operating voltages applied to this device present a shock hazard.

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